PHONETICALLY-DRIVEN EPENTHESIS ASYMMETRIES IN FRENCH AND SPANISH OBSTRUENT-LIQUID CLUSTERS

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1. *Introduction*

It has been argued that obstruent-liquid (OL) clusters constitute the optimal branching onset (Clements 1990, among others). As concerns French and Spanish in particular, these clusters have been the object of various analyses, which have focused on their phonology (e.g. Dell 1995; Bradley 1999); their cross-dialectal realization (e.g. Malmberg 1965; Bradley 1999); and their acquisition (e.g. Rose 2000; Steele 2002; Barlow 2003). In spite of this considerable body of research, there are surprisingly few laboratory studies which detail the phonetic realization of such clusters. For this reason, we present the results of an acoustic analysis of OL clusters in Quebec French and Argentinean Spanish. Particular attention will be paid to the patterns of cluster-medial vocalic epenthesis observed.

Examples of the French and Spanish clusters under investigation are given in (1a,b).

(1)	a. <u>French</u>		
	précis	[pʁesi]	"precise"
	gifler	[ʒifle]	"to slap"
	b. <u>Spanish</u>		
	prevé	[preve]	"s/he foresees"
	aflora	[aflora]	"it emerges"

Under standard phonological analyses, such clusters constitute branching onsets (i.e. [ʒi.fle], [a.flo.ɾa]). While French and Spanish OL clusters are phonologically similar in their syllabification, they differ in at least two respects as concerns their phonetics. First, whereas French <r> is realized as the voiced

uvular fricative [ʁ],[1] the Spanish rhotic is the voiced alveolar tap [ɾ]. Second, a number of researchers (Malmberg 1965; Quilis 1970, 1993; Ramírez 2002) have noted the presence of an epenthetic vowel breaking up the OL cluster in Spanish. To our knowledge, no such observation has been made for French.

In this paper, we investigate this apparent between-language epenthesis asymmetry using experimental data from Quebec French and Argentinean Spanish. In doing so, we set out two goals. First, we will demonstrate that cluster-medial epenthesis is not limited to Spanish but also occurs in French, although in a more restricted set of environments. Second, following others (e.g. Malmberg 1965; Ohala 1992), we will argue that the epenthesis attested is a form of dissimilation which serves to increase the articulatory distance between the two members of the cluster. We will show that distance can be measured on the manner, place and voicing dimensions, with voicing being the most important of the three. On the manner dimension, continuancy plays a crucial role in both languages. In particular, if the two members of the cluster are [-continuant], epenthesis occurs quasi-categorically. As for place, epenthesis tends to be more frequent when both members of the cluster share place features. Finally, in French, epenthesis occurs only with clusters containing voiced obstruents; in Spanish, epenthetic vowels are significantly longer in these environments.

The rest of the paper is structured as follows. In §2, we outline the basic phonological and phonetic properties of French and Spanish OL clusters. We attempt to explain the epenthesis previously reported for Spanish and also attested in the current French data in the context of the dissimilation hypothesis in §3, where we forward a number of hypotheses. The experimental study used to test these hypotheses is outlined in §4, followed by the reporting of the results in §5. In §6, we evaluate each of the hypotheses, providing explanations for those cases refuted by the data. We conclude briefly in §7.

2. *Obstruent–liquid clusters in French and Spanish*

In the following sections, we provide a characterization of the basic phonological and phonetic properties of French and Spanish OL[2] clusters relevant to the present work.

[1] French <r> is standardly described as uvular (e.g. Léon 1992). As concerns Quebec French, the variety to be investigated here, both O'Shaughnessy (1982:383) and Tousignant (1987:75) describe it as velar. The exact place of articulation is not relevant to the current discussion, only that it be dorsal.

[2] We use the term 'liquid' here to designate a subset of laterals and rhotics, namely [l,ʁ] in French and [l,ɾ] in Spanish. See Colantoni & Steele (2005) for further discussion of this term.

2.1 *French OL clusters*

While French permits a variety of cluster types, OL clusters are the most frequent and varied. Such clusters occur in all positions within the word (initial: *travail* [tʁavaj] "work"; medial: *conclure* [kɔ̃klyʁ] "to conclude"; final: *vinaigre* [vinɛgʁ] 'vinegar'). For both lateral and rhotic clusters, all permutations of obstruent manner, place and voicing are permitted with one exception: clusters consisting of a coronal obstruent plus lateral are illicit (e.g. *tlaie [tle]; *dlore [dlɔʁ]), except when syllabified heterosyllabically (e.g. *athlète* [at.lɛt] "athlete"; *modeler* [mɔd.le] "to model").

As stated earlier, we are unaware of any reports of cluster-medial epenthesis in French. Indeed, phonetic descriptions of these clusters typically describe the direct transition from the obstruent to the liquid (e.g. Delattre 1966: 115; O'Shaughnessy 1982: 390-391; Léon 1992: 71). This is also true for OL clusters derived via schwa deletion, both as concerns production and perception (Fougeron & Steriade 1997; Fougeron & Steriade 1999).

2.2 *Spanish OL clusters*

OL clusters are one of the few consonant combinations allowed in Spanish. Phonologically, they syllabify as branching onsets. The number of possible permutations of obstruent and liquid is similar to that observed in French.[3] Indeed, most combinations of manner, place and voicing are possible, with the exception of voiced coronal-lateral sequences. Voiceless coronal-lateral onset clusters are permitted in some varieties, including Argentinean Spanish, but are syllabified heterosyllabically in most others. As concerns position in the word, Spanish OL clusters may appear initially or medially (e.g. *drama* [dɾa.ma] "drama"; *agrio* [a.ɣɾjo]] "sour").

In contrast to French, previous experimental studies have indeed shown that Spanish OL sequences may be broken up by a cluster-medial epenthetic vowel (Malmberg 1965; Quilis 1970, 1993; Ramírez 2002), as illustrated in (2).[4]

(2) *prevé* /pɾeve/ → [pəɾeve] "s/he foresees"
sopla /sopla/ → [sopəla] "s/he blows"

[3] For the existing accidental gaps, see the table of Spanish experimental stimuli in the Appendix.

[4] This contrasts with native speaker intuitions. For example, following their participation in the present study, subjects were informally asked if they were aware of the cluster-medial vowel and, without exception, indicated not being so.

This research has consistently observed that the epenthetic vowel can be as long as an unstressed vowel, with a mean length of approximately 32ms (Quilis 1970:143). The quality of the epenthetic vowel varies depending on that of the surrounding vowels, although authors mostly agree that it copies the quality of the following vowel (Quilis 1970:143). Epenthetic vowels have been consistently reported for obstruent-rhotic clusters, less so for obstruent-lateral clusters (see Ramírez 2002). Previous studies, however, do not explore the motivations of this asymmetry, nor do they analyze in sufficient detail the role that similarities in manner, place and voicing between the obstruent and liquid play in predicting the cross-dialectal patterns observed.

In the remainder of the present study, we will demonstrate that OL cluster-medial epenthesis is productive in both French and Spanish and that the rate of epenthesis, as well as the length of the epenthetic vowel, are a function of the similarity between the two members of the cluster; that is, the more similar the members of the cluster, the higher the rate of epenthesis. In this way, epenthesis is interpreted as a type of dissimilation. We turn to the dissimilation hypothesis in the following section.

3. *The dissimilation hypothesis*

Syllables with branching onsets are typologically less common than CV syllables, and can be simplified via three alternative strategies, all of them attested in Romance languages diachronically and/or synchronically (Walker 1984; Lloyd 1993; Penny 2002, among others). These include: (i) deletion of one member of the cluster; (ii) assimilation; and/or (iii) epenthesis. Only assimilation and epenthesis are observed in the present data. Assimilation has been reported for clusters involving obstruents and laterals, both synchronically (Gibbon et al. 1993; Romero 1996) and diachronically (Lloyd 1993; Penny 2002). In the current study, as we will discuss below, assimilation is also observed in French obstruent-rhotic clusters. In general, assimilation only takes place when the second member of the cluster is [+continuant].

We will focus, however, on the most frequently used strategy for cluster-simplification, namely vowel epenthesis. Several authors (Malmberg 1965; Ohala 1992) have argued that epenthesis is a type of dissimilation, i.e., a strategy to increase the articulatory and perceptual distance between two segments. On this view, we expect that increased similarity between the obstruent and liquid will lead to a higher rate of epenthesis. Similarity, according to some authors, is not categorical but gradient, being dependent upon the number of features shared by two or more segments (e.g. Pierrehumbert 1992;

Frisch et al. 1997): the greater the number of overlapping features, the stronger the tendency towards dissimilation.[5]

The importance of featural contrast for onset cluster well-formedness is witnessed to by the fact that the unmarked complex onset is a stop-liquid sequence, that is a low-sonority non-continuant followed by high-sonority continuant. When this ideal segmental profile is not met – for example, if one of the members is either non-continuant or the sonority difference is insufficient – the likelihood of simplification increases, with the increase being directly proportional to the number of features shared. To illustrate, a higher rate of epenthesis should be expected when the two members of the clusters are non-continuant, as is the case with Spanish obstruent-rhotic clusters.[6] We argue here that manner plays a crucial role in epenthesis, since a necessary (although insufficient) condition for epenthesis to take place, is the presence of at least one non-continuant segment in the cluster. In those cases where the cluster consists of two continuant segments (i.e. the more marked case), assimilation occurs.

Voicing and manner interact with the place dimension. There is a cross-linguistic tendency to prohibit OL sequences with identical place (Rice 1992, among others), i.e. clusters such as [tl], [dl] are dispreferred. Note, however, that this tendency is gradient. For example, while [dl] clusters, which agree in both voicing and place, are categorically disallowed in Spanish, [tl] clusters, which differ in voicing but not place, are allowed in some varieties (e.g. both Mexican and Argentinean Spanish).

3.1 *Predicted asymmetries*

The asymmetries to be discussed here follow from our general proposal that the similarity between the members of a cluster determines the particular phonetic realization. Specifically, the greater the degree of identity between the obstruent and the liquid in manner, place of articulation and voicing, the greater the tendency towards simplification, either via vowel epenthesis or cluster reduction (e.g. affrication). Accordingly, we predict the three asymmetries outlined in the following sections.

[5] We assume the following features: manner [±continuant]; place [labial, coronal, dorsal]; and voicing [±voice]. Ultimately, what is critical is not the exact features, but rather their similarity/difference.

[6] As such, epenthesis is also expected in environments where spirantization occurs.

3.1.1 *Manner of articulation.* In both French and Spanish, the lateral is a sonorant, continuant[7] segment. In contrast, the rhotic is either non-sonorant (French /ʁ/) or non-continuant (Spanish /ɾ/).[8] As such, laterals are phonetically less similar to obstruents than rhotics in both languages and, consequently, less dissimilation, specifically a lower rate of epenthesis, should be expected in obstruent-lateral clusters. Moreover, a higher rate of epenthesis is predicted for Spanish, since obstruent-rhotic clusters consist of two non-continuant segments.

This prediction is supported by diachronic, cross-dialectal and experimental evidence in Spanish. First, obstruent-lateral clusters evolved differently from Latin vis-à-vis their obstruent-rhotic counterparts. While the former underwent assimilation and eventually palatalization, the latter exhibited a greater tendency towards maintenance (Lloyd 1994; Penny 2002). Second, cross-dialectal studies reveal different assibilation patterns for obstruent-lateral versus obstruent-rhotic clusters (Lenz 1940; Alonso 1953; Lipski 1994; Penny 2000, among others); specifically, the former may undergo assibilation while the latter do not. Finally, articulatory studies on Spanish and Catalan have observed assimilation in both lateral-obstruent clusters (Romero 1996) and obstruent-lateral clusters (Recasens & Pallarès 2001). In contrast, sequencing is reported for rhotic-obstruent (Romero 1996) and obstruent-rhotic clusters (Recasens & Pallarès 2001).

3.1.2 *Place of articulation*. We predict a between-language asymmetry related to the obstruent's place of articulation. In French, the rhotic is dorsal; thus, dorsal obstruent-rhotic clusters should involve greater simplification (i.e. epenthesis or reduction). In Spanish, where the rhotic is alveolar, more incidences of simplification should be observed with coronal obstruents.[9] This

[7] Some authors (Cressey 1978; Núñez Cedeño & Morales Front 1998) characterize Spanish laterals as [-continuant] and rhotics as [+continuant]. We will follow Quilis (1993) and Martínez Celdrán (1994) here, since acoustically, laterals are periodic sounds, while the rhotic tap is a short stop.

[8] French rhotics are traditionally described as non-sonorant. However, acoustic analyses (e.g. O'Shaughnessy 1982:390) demonstrate that they may be realized as approximants (i.e. as sonorant segments). As concerns Spanish, in the variety under study, <r> is always realized as a tap (i.e. as [-continuant]) (see Borzone de Manrique 1980). In other dialects, it may be realized as a fricative (Lipski 1994).

[9] Both Spanish liquids are traditionally described as alveolar (Navarro Tomás 1970). However, recent phonological and phonetic studies have argued against an identical place of articulation for /l/ and /ɾ/ (Colantoni 2001; Recasens & Pallarès 2001). According to these analyses, laterals are front alveolars as opposed to rhotics, which are back alveolars (Recasens & Pallarès 2001: 288).

prediction is supported by cross-dialectal assibilation asymmetries, with assibilation in clusters being more frequent in coronal-rhotic sequences (Malmberg 1965; Navarro Tomás 1970; Bradley 1999).

3.1.3 *Voicing*. If the segments agree in voicing, we expect higher rates of epenthesis or reduction. Again, the prediction is supported by diachronic and synchronic variation in Spanish. Clusters in which both members agreed in voicing evolved differently from Latin than clusters with different voicing specifications (Lloyd 1993; Penny 2002). Finally, clusters that agree in voicing show lower rates of assibilation (Malmberg 1965).

4. ***Current study***

In order to test our hypotheses, 10 speakers (5 male; 5 female) of Quebec French, and 11 speakers (5 male; 6 females) of Buenos Aires Spanish were recorded. These varieties were chosen both for pragmatic reasons - availability of speakers - and, in the case of Spanish, to avoid the possibility that the OL clusters under investigation be realized as affricates (e.g. Chilean Spanish; Alonso 1953). The stimuli used consisted of 48 words containing the target OL clusters (see Appendix), as well as 27 distractors. The stimuli were controlled for (i) the type of liquid; (ii) the manner, place and voicing of the obstruent;[10] and (iii) the position of the cluster in the word, including stress.

The data were elicited via a word-reading task in which the targets and distractors were intermixed. All words were presented in the same carrier sentence (French: “Je dis [TARGET] encore une fois”; Spanish: “Digo [TARGET] otra vez”; ‘I say [TARGET] again’) in random order three times, generating 134 (48 x 3) tokens per subject.[11] Testing sessions were recorded using a Marantz CDR300 CD recorder (44100Hz; 32bit; Stereo) and a unidirectional Audiotechnica AT803B lavaliere.

[10] In order to have comparable sets of stimuli, no /v/-initial clusters were included, given their absence in Spanish.

[11] The reason for placing the targets in carrier sentences was to permit measurement of the stop. One consequence, however, was the increased probability of underlying voiced stops being spirantized (i.e. realized as approximants). Indeed, in both languages, some speakers did exactly this, with a higher rate observed in Spanish. Note that such tokens were grouped with other clusters for which the underlying initial consonant was a stop in the statistics. Standard phonological descriptions for Spanish (Navarro Tomás 1970; Quilis 1993) claim that spirantization is a categorical process. The patterns in our data demonstrate that it is a variable (gradient, non-categorical) phenomenon. In order to determine whether spirantization has an effect on the length of the epenthetic vowel, a speaker-by-speaker analysis is needed; we leave this task for future studies.

The sound files were downsampled (22050Hz; 16-bit; Mono), and low-pass filtered (11050Hz). Target words were then extracted and labelled. The clusters were analyzed with PRAAT 4.0.41 for (i) the presence/absence of the epenthetic vowel; (ii) the length and first three formants of the vowel, when present; and (iii) voicing. Results were entered in a spread-sheet, and statistics were calculated with SAS 8.2. The confidence level for all the statistics was set at 0.05.

5. ***Results***

As shown in Table 1, epenthetic vowels are detected in both French and Spanish. The rate of epenthesis varies across liquid types, obstruent manner and voicing, as well as between languages, with the exception of obstruent-lateral clusters. In the case of these latter clusters, epenthetic vowels are almost categorically absent, with an epenthesis rate of less than 2% in both languages. In all other cases, epenthetic vowels, when present, have a mean intensity and length similar to that of unstressed vowels (see waveform in Figure 1).

		French			**Spanish**		
	Cluster	N	Total	%	N	Total	%
<**l**>	Obstruent + /l/	9	599	**1.5**	11	591	**1.9**
<**r**>	/fr/	3	122	**2.5**	168	198	**84.9**
	/p,t,k/ + /r/	20	356	**5.6**	339	360	**94.2**
	/b,d,g,/ + /r/	320	356	**89.9**	386	395	**97.7**
		352	1433		922	1544	

Table 1. *Rate of epenthesis in French and Spanish*

We now turn to the particular phonetic realization of these vowels in both languages.

5.1 *Epenthesis in French*

In French, epenthetic vowels are observed for all speakers almost exclusively in voiced obstruent-rhotic clusters,[12] an example of which is given in Figure 1. There is an effect for place, with the rate of epenthesis being significantly lower with dorsals (85%; χ^2= 4.94, p<.05) than coronals (93.2%) or labials (91.5%).

[12] There is only one speaker in the sample who epenthesizes in both voiced and voiceless rhotic clusters. However, the rate of epenthesis is higher with voiced obstruents.

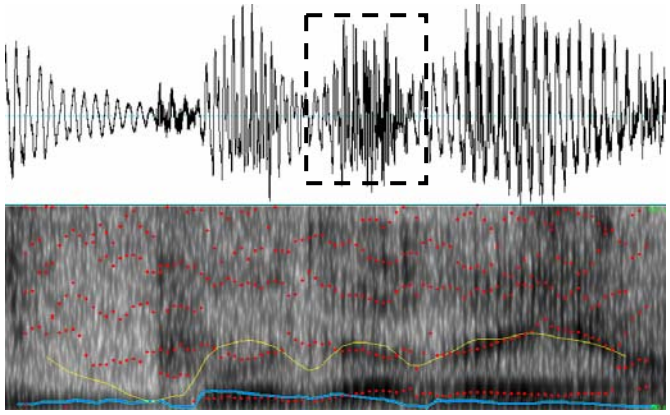

Figure 1: *Epenthetic vowel in French voiced obstruent-rhotic cluster (target poudrer /pudʁe/ 'to powder')*

The mean duration of the vowel varies across speakers. Given that our subject group included similar numbers of males and females, statistics were run to determine the possibility of a gender-based difference; no effect was found. Rather, differences seem to be an individual characteristic of the speaker. 8 out of 10 of the speakers in the sample constitute a group with no significant difference in the length of their epenthetic vowel, which may range from 27ms to 38ms; this is comparable to Quilis' (1970) findings for epenthetic vowels in Spanish (see §2.2).[13] The other two speakers, one male and one female, produce a significantly longer vowel in the range of 50ms. The length of the epenthetic vowel is affected by the place of the obstruent: it is significantly longer in coronal clusters ($F_{(3, 39)}$=3.26, p=.03), than in labial or dorsal clusters, between which there was no significant difference in vowel length. Finally, the duration of the epenthetic vowel is not significantly affected by the position of the cluster in the word (initial vs. internal; t= 0.86; p>.38) or by the presence versus absence of stress (t= 1.66; p>.09).

As concerns the quality of the epenthetic vowel, it is affected by the preceding (F1: $F_{(4, 356)}$=7.12, p=.0000; F2: $F_{(4, 356)}$=15.21, p=.0000) and following vowels (F1: $F_{(4, 356)}$ =19.87, p=.0000; F2: $F_{(4, 356)}$=9.35, p=.0000).[14]

[13] Subjects were instructed to read at a normal speech rate. With the exception of one speaker, this directive was respected. Indeed, the length of all rounds was similar across speakers (i.e. approximately 2.5 minutes).

[14] Previous studies for Spanish (e.g. Quilis 1970) report that the epenthetic vowel had the same quality as the following vowel. In order to test whether this was the case for French and Spanish here, we measured the formant frequencies and ran statistics to test the role of the flanking vowels. In contrast to Quilis (1970), we found an effect for both vowels. Since the

5.2 *Epenthesis in Spanish*

The Spanish data display a greater overall rate of epenthesis than those from French. Indeed, epenthetic vowels are attested in both voiceless and voiced obstruent-rhotic clusters, as shown in Figure 2 (a, b), and were detected for all the speakers in the sample. Note that epenthesis in fricative-rhotic clusters is highly variable, with the rate ranging from 27% to 100% among speakers. As concerns the effect of place, there is virtually no difference between coronal (96.6%), labial (95.4%), and dorsal (94.9%) clusters.

As with French, the obstruent's place of articulation also plays a role in determining the length of the epenthetic vowel; the vowel is significantly longer in dorsal clusters ($F_{(2,\ 743)}$= 9.3, p=0.0001). Voicing also influences length, with longer vowels found in voiced clusters (t(936)= 14.99, p<0.0000). In contrast to French, the position of the cluster in the word and the presence of stress on the following vowel do condition epenthesis in Spanish. The epenthetic vowel is significantly longer in word-internal versus word-initial clusters (t(936)=3.55; p<0.0004), and in stressed versus unstressed syllables (t(936)=3.52, p<0.0004). For 8 of the 11 speakers, there is no significant difference in the mean length of the vowel which ranges from 31ms to 40ms; two speakers (one male and female) have a significantly lower mean around 20ms, and one of the speakers has a significantly higher mean, around 47ms. Again, as in French, gender plays no role in determining the length of the epenthetic vowel.

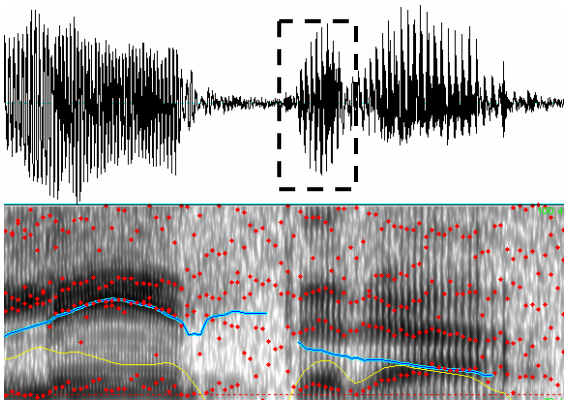

Figure 2a: *Epenthetic vowel in Spanish voiceless obstruent-rhotic cluster (target letra /letɾa/ 'letter')*

objective of the present paper is not to discuss the quality but rather the presence and length of the vowel, no further tests were performed.

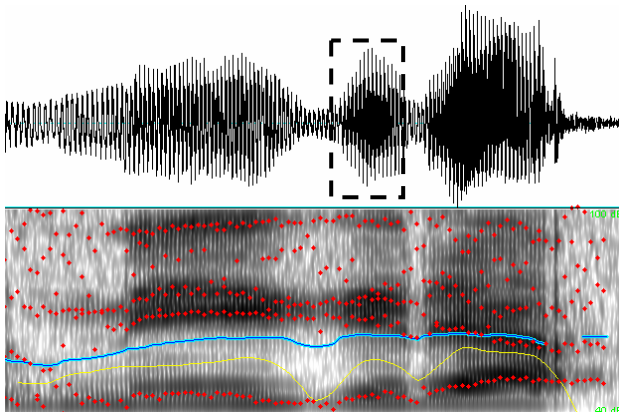

Figure 2b: *Epenthetic vowel in Spanish voiced obstruent-rhotic cluster (target negra /neɣra/ 'black')*

The quality of the epenthetic vowel is significantly affected by the preceding (F1: $F_{(4, 953)}$=13.25, p=.0000; F2: $F_{(4, 953)}$=16.81, p=.0000) and following vowels (F1: $F_{(4, 953)}$ =36.98, p=.0000; F2: $F_{(4, 953)}$=13.13, p=.0000).

5.3 *Summary*

The experimental data from the present study witness to three asymmetries in the phonetic realization of French and Spanish OL clusters. The first asymmetry exists between lateral and rhotic clusters. In both languages, virtually no epenthesis is observed in lateral clusters, while there is variable epenthesis in rhotic clusters. The second asymmetry, affecting only rhotic clusters, involves voicing. In French, epenthesis is attested almost exclusively in voiced obstruent-rhotic clusters. In contrast, all Spanish rhotic tokens contain an epenthetic vowel, but the epenthetic vowel is significantly longer in voiced clusters. The final asymmetry involves place of articulation: French epenthetic vowels are longer in coronal clusters, whereas Spanish epenthetic vowels are longer in dorsal clusters.

6. ***Analysis and discussion***

To this point, we have demonstrated that cluster-medial epenthesis in OL clusters is not restricted to Spanish. Moreover, we have shown that there exist three asymmetries in the phonetic realization of such clusters. In the remainder of this article, we will seek to explain these asymmetries in the context of the dissimilation hypothesis. We begin by evaluating the three hypotheses forwarded in §3.1.

6.1 *Hypothesis evaluation*

In §3, we predicted that the rate of epenthesis would be a function of the similarity between the members of the cluster, i.e. the more similar the two consonants in terms of manner, place or voicing, the higher the rate of epenthesis, and possibly the longer the epenthetic vowel. We look at each of the specific hypotheses in turn.

6.1.1 *Manner of articulation.* We hypothesized that, as laterals are phonetically less similar to obstruents than rhotics on either the sonority (French) or continuancy (Spanish) dimension, a lower rate of epenthesis should be expected in obstruent-lateral clusters in both languages. In addition, we predicted a lower rate of epenthesis in French, where the rhotic is continuant.

Our results support this hypothesis. Virtually no epenthesis (under 2%) was observed in obstruent-lateral clusters, in contrast to the high rate of epenthesis observed in obstruent-rhotic clusters. The rate with rhotics was higher in Spanish (94%), where epenthesis is attested in both voiceless and voiced clusters, yet nonetheless relatively high in French at 41%. A closer look at the data reveals that the lower rate of epenthesis in French is the result of competing strategies of cluster simplification. Recall from §3 that epenthesis is but one of several dissimilatory strategies available for cluster simplification. An alternative tactic involves assimilation, including the reduction of the cluster via affrication. While French voiceless rhotic clusters differed from those of Spanish in generally not triggering epenthesis, they often involved such simplification. For the group, 30% of voiceless labial-rhotic clusters were blended; the rate increased to 49% and 71% for voiceless coronal and dorsal tokens respectively. Assimilation in voiceless rhotic clusters is not surprising, particularly for clusters composed of a dorsal obstruent, which shares the place of articulation with /ʁ/.

6.1.2 *Place of articulation.* Following the dissimilation hypothesis, we predicted that the rate of epenthesis would be higher in clusters that shared place of articulation; this is not supported. In French, where rhotic /ʁ/ is dorsal, we predicted higher rates in /kʁ/ and /gʁ/ clusters. However, epenthesis in our data is disfavoured in dorsals; indeed, it is significantly lower (85%) than in coronals (93%) and labials (91%). In Spanish, where the rhotic /ɾ/ is coronal, we predicted greater epenthesis in coronal clusters. Yet, there is no significant difference across places of articulation (coronal: 97%; labial and dorsal: 95%).

If we look at the length of the epenthetic vowel, we observe another asymmetry in both languages that goes against the prediction of greater dissimilation in clusters where the obstruent and liquid share their place of articulation; namely, epenthetic vowels are significantly longer in French

coronal clusters, while the epenthetic vowel in Spanish is significantly longer in clusters with a dorsal obstruent. This asymmetry is not without explanation if one takes into account the phonetic characteristics of the rhotics in both languages. In French, the rhotic is dorsal, while in Spanish it is coronal. As a result, rhotic clusters consist of a coronal-dorsal sequence or the mirror dorsal-coronal sequence. In both cases, the production of the clusters involves tongue displacement, with the gestural adjustment leading to longer epenthetic vowels. Such displacement is absent from labial-rhotic clusters, and from clusters with identical place of articulation.

6.1.3 *Voicing*. It was hypothesized that the rate of epenthesis would be higher in clusters that agreed in voicing. This third prediction is supported by the French and Spanish data. In French, there is a clear voicing asymmetry in rhotic clusters favouring epenthesis in voiced tokens. In Spanish, although epenthetic vowels are observed in both voiced and voiceless clusters, they are significantly longer in the former. Thus, voicing plays the most important role in cluster simplification.

The observation that voicing has the strongest effect in cluster simplification deserves further explanation. We hypothesize that it is not the presence versus absence of vocal fold vibrations that is relevant, but that voiced obstruents are shorter than their voiceless counterparts (Colantoni & Steele 2004). Thus, the presence of the epenthetic vowel, in French, and the longer epenthetic vowel, in Spanish, would be a compensatory effect to preserve isochrony.

6.2 *Phonetics or phonology?*

One final question which we wish to address is the nature of the epenthesis attested in the present data. Specifically, are the French and Spanish cluster-medial vowels driven by the phonetics or phonology?

We believe that what we are observing is primarily a phonetic phenomenon. While many of the French and Spanish epenthetic vowels resemble underlying vowels in both length and intensity, they do not appear to participate in phonological phenomena. For French, word-final syllabification constitutes such a phenomenon. Consider a form such as *sucre* /sykʁ/ “sugar”. In Quebec French, syllabification of such a form involves deletion of the liquid, resulting in [syk]. Simplification arguably occurs because word-final OL clusters, highly marked sequences cross-linguistically, are illicit in this variety. Deletion allows for a surface form attested elsewhere in the language (compare [syk] and *duc* [dyk] “duke”). Now consider a voiced final OL cluster, such as that of *maigre* “thin”. If the medial vowel observed in voiced obstruent-rhotic clusters in the present Quebec French data were phonological

in nature, learners' underlying representation of *maigre* would be /mɛgəʁ/. One would thus expect a syllabification of [mɛ.gəʁ], on par with forms such as *bancaire* /bɑ̃kɛʁ/→[bɑ̃.kɛʁ] 'banking'. However, syllabifications such as [mɛgəʁ] are unattested. Indeed, word-final voiceless and voiced obstruent-rhotic clusters pattern identically as concerns their syllabification; *maigre* can only be realized as [mɛg].

In Spanish, one might use stress as a diagnostic (see e.g. Bradley 2004). Were the Spanish cluster-medial vowels underlying, they should be stressed when syllabified in the antepenultimate syllable of word with a light penult. For example, in the case of a form such as *tímbrico* in (3), stress would fall on the second syllable [bə], were the vowel underlying. However, as shown in (3b), stress falls on the word-initial syllable.

(3) a. [tim.ˈbə.ɾi.ko] "timbral"
b. [ˈtim.bə.ɾi.ko]

Note that, in order to use stress as a diagnostic, one requires words of three or more syllables containing an obstruent-rhotic cluster in the penult. Such words are few in number and are generally technical or learned vocabulary. As such, the stress patterns of these words could be learned on a word-by-word basis. Were this the case, stress placement in such words would not be an appropriate diagnostic.

Keeping this potential limitation in mind, we cautiously forward the syllabification of final OL clusters in French and the Spanish stress pattern in (3) as evidence for the phonetic nature of the cluster-medial vowels observed in the current data. Determining whether or not the epenthetic vowels described here have either synchronic or diachronic consequences for the phonology of either or both languages will be left for future research.

7. *Conclusions*

In this paper, we have presented data from an experimental study of French and Spanish OL clusters that demonstrate the productivity of cluster-medial epenthesis in both languages. We have shown that the degree of simplification via epenthesis is a function both of the type of liquid and of the similarity of the members of the cluster in manner, place and voicing. Of these four dimensions, liquid type, followed by voicing, has the strongest effect.

The tendency towards cluster simplification observed is consistent with both a general Romance and cross-linguistic trend favouring maintenance of clusters exhibiting greater sonority contrast. Moreover, the present results indicate that both French and Spanish observe the Romance tendency towards an

asymmetrical evolution of (i) obstruent-lateral and obstruent-rhotic clusters, and (ii) voiced versus voiceless obstruent clusters.

The findings of this paper are limited to the production of OL clusters in Quebec French and Argentinean Spanish. In current work, we are studying the realization of these clusters in two other varieties, namely European (France) French and Chilean Spanish. Preliminary analysis of the data reveals the presence of epenthetic vowels in the same environments described here; further analysis will be required to determine the effect of the other variables (e.g. voicing, stress) on the realization of these vowels. In future research, we also intend to study the perception of these clusters. As stated earlier, native speakers report being unaware of the epenthetic vowel when questioned. However, this does not necessarily mean that the vowel is not perceived at a prelexical level. If this is the case, it is possible that phonetic epenthesis may have long-term effects on the phonology of OL clusters in French and Spanish.

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Appendix: Experimental Stimuli

French

Word shape	*Liq*	*Labial* p	b	f	*Coronal* t	d	*Dorsal* k	g
CLV	[l]	*plat* [plɑ]	*blé* [ble]	*flot* [flo]	*	*	*clé* [kle]	*glas* [glɑ]
	[ʁ]	*pré* [pʁe]	*bras* [bʁɑ]	*frais* [fʁɛ]	*très* [tʁɛ]	*drap* [dʁɑ]	*crée* [kʁe]	*gras* [gʁɑ]
CLV.'CV	[l]	*placer* [plase]	*blaguer* [blage]	*flatter* [flate]	*	*	*classer* [klase]	*glacé* [glase]
	[ʁ]	*précis* [pʁesi]	*bravo* [bʁavo]	*frapper* [fʁape]	*tracer* [tʁase]	*drapeau* [dʁapo]	*craquer* [kʁake]	*gratter* [gʁate]
CV.'CLV	[l]	*couplet* [kuple]	*doubler* [duble]	*siffler* [sifle]	*	*	*boucler* [bukle]	*régler* [ʁegle]
	[ʁ]	*mépris* [mepʁi]	*cobra* [kobʁɑ]	*coffret* [kɔfʁɛ]	*vitré* [vitʁe]	*poudrer* [pudʁe]	*sucré* [sukʁe]	*degré* [dəgʁe]
CV. CLV.'CV	[l]	*déplacer* [deplase]	*déblayer* [debleje]	*refléter* [ʁeflete]	*	*	*déclarer* [deklaʁe]	*déglacer* [deglase]
	[ʁ]	*soprano* [sopʁano]	*librairie* [libʁeʁi]	*naufragé* [nofʁaʒe]	*détraquer* [detʁake]	*redresser* [ʁədʁese]	*décrasser* [dekʁase]	*dégrader* [degʁade]

Appendix: Experimental Stimuli (cont'd)

Spanish

Word shape	*Liq*	*Labial* p	b	f	*Coronal* t	d	*Dorsal* k	g
CLV	[l]	*plan* [plan]	*bla* [bla]	*flor* [flor]	*	*	*clan* [klan]	*
	[ɾ]	*	*	*frac* [frak]	*tras* [tras]	*	*	*gran* [gran]
CLV. 'CV	[l]	*plegó* [pleɣo]	*blasón* [blason]	*floté* [flote]	*	*	*clavé* [klaβe]	*glace* [glase]
	[ɾ]	*prevé* [preβe]	*bramó* [bramo]	*frenó* [freno]	*traté* [trate]	*drogó* [droɣo]	*creyó* [kreʒo]	*grabé* [graβe]
CV. 'CLV	[l]	*soplé* [sople]	*doblá* [doβla]	*	*	*	*	*
	[ɾ]	*	*cobrá* [koβra]	*sufrí* [sufri]	*	*podré* [poðre]	*lucrá* [lukra]	*lograá* [loɣra]
'CV.CLV	[l]	*sopla* [sopla]	*tabla* [taβla]	*	*	*	*tecla* [tekla]	*regla* [reɣla]
	[ɾ]	*lepra* [lepra]	*sobra* [soβra]	*cifra* [sifra]	*letra* [letra]	*sidra* [siðra]	*sacra* [sakra]	*negra* [neɣra]
'CLV.CV	[l]	*plato* [plato]	*bledo* [bleðo]	*flaco* [flako]	*	*	*claro* [klaro]	*globo* [gloβo]
	[ɾ]	*prado* [praðo]	*brazo* [braso]	*frase* [frase]	*traje* [traxe]	*drama* [drama]	*crema* [krema]	*grave* [graβe]